

2. (10 points) Linear Time-invariant Systems

a. (5 points) An LTI system has the impulse response $h(t) = e^{-t} \cos(t) u(t)$. Is this system BIBO stable?

Calculate the response $y(t)$ to the input signal $x(t) = t u(t)$.

$H(s) = \frac{s+1}{(s+1)^2 + 1}$, $X(s) = \frac{1}{s^2} \Rightarrow Y(s) = \frac{(s+1)}{(s+1)^2 + 1} \cdot \frac{1}{s^2}$
 poles: $\{-1 \pm j\}$
 $\Rightarrow y(t) = \frac{1}{2} t u(t) - \frac{1}{2} e^{-t} \sin(t) u(t)$
 BIBO STABLE
 Grading: 2 pts each (A), (B), +1 pt (C)

b. (5 points) Calculate the response $v_{out}(t)$ of the following system to the input $v_s(t) = x(t)$.

$x(t) = \sum_{n=1, n=\text{odd}}^{\infty} \frac{8A}{n^2 \pi^2} \sin\left(\frac{n\pi}{2}\right) \sin\left(\frac{2n\pi t}{T_0}\right)$
 $\hat{H}(\omega) = \frac{j\omega L}{R + j\omega L}$
 $|\hat{H}(\omega)| = \frac{\omega L}{\sqrt{R^2 + (\omega L)^2}}$
 $\phi(\omega) = \pi/2 - \tan^{-1}\left(\frac{\omega L}{R}\right)$
 $\omega_0 = \frac{2\pi}{T_0}$
 Grading: 1 pt for trying, +2 pts each (A) & (B)

Why is the amplitude missing the "L" in it?

2a

$$\mathcal{L}[h(t)] = \frac{s+1}{(s+1)^2 + 1}, \quad \mathcal{L}[x(t)] = \frac{1}{s^2}$$

$$\frac{s+1}{(s+1)^2 + 1} \times \frac{1}{s^2} = \frac{s+1}{s^2((s+1)^2 + 1)}$$

Set factors to zero and solve for s:

$$s^2 = 0 \rightarrow s = 0$$

$$(s+1)^2 + 1 = 0 \rightarrow (s+1)^2 = -1 \rightarrow s+1 = \sqrt{-1} = \pm j \rightarrow s = -1 \pm j$$

The -1 shifts the imaginary pole of the imaginary axis to the left, making the system **Stable**.

Response to input calculation

$$\left(\frac{s+1}{s^2((s+1)^2 + 1)} = \frac{A}{s} + \frac{B}{s^2} + \frac{Cs+D}{(s+1)^2 + 1} \right) s^2((s+1)^2 + 1)$$

$$s+1 = A(s((s+1)^2 + 1)) + B((s+1)^2 + 1) + (Cs+D)s^2$$

$$\rightarrow s((s^2 + 2s + 1) + 1) = s(s^2 + 2s + 2) = s^3 + 2s^2 + 2s$$

$$\rightarrow s+1 = A(s^3 + 2s^2 + 2s) + B(s^2 + 2s + 2) + (Cs+D)s^2$$

$$\rightarrow s+1 = As^3 + 2As^2 + 2As + Bs^2 + 2Bs + 2B + Cs^2 + Ds^2$$

$$\rightarrow s+1 = 1s+1 \rightarrow 0s^3 + 0s^2 + 1s+1$$

$$\rightarrow s^3(A+C) + s^2(2A+B+D) + s(2A+2B) + s^0(2B)$$

$$0s^3 + 0s^2 + 1s+1 = s^3(A+C) + s^2(2A+B+D) + s(2A+2B) + s^0(2B)$$

Solve constants by setting them equal to their respective powers

$$\bullet 1 \& s^0(2B) \sim 1 = 2B \rightarrow B = \frac{1}{2}$$

$$\bullet 1s \& s(2A+2B) \sim 1 = 2A + 2\left(\frac{1}{2}\right) \rightarrow 2A = 0 \rightarrow A = 0$$

$$\bullet 0s^2 \& s^2(2A+B+D) \sim 0 = 2(0) + \frac{1}{2} + D \rightarrow D = -\frac{1}{2}$$

$$\bullet 0s^3 \& s^3(A+C) \sim 0 = (0) + C \rightarrow C = 0$$

Plug Back in, and $\mathcal{L}^{-1}$

$$\frac{A}{s} + \frac{B}{s^2} + \frac{Cs+D}{(s+1)^2 + 1} \rightarrow \frac{0}{s} + \frac{(\frac{1}{2})}{s^2} + \frac{(-\frac{1}{2})}{(s+1)^2 + 1}$$

$$\mathcal{L}^{-1}\left[\frac{1/2}{s^2}\right] = \frac{1}{2} t u(t)$$

$$\mathcal{L}^{-1}\left[\frac{-1/2}{(s+1)^2 + 1}\right] = -\frac{1}{2} e^{-t} \sin(t) u(t)$$

$$y(t) = \frac{1}{2} t u(t) - \frac{1}{2} e^{-t} \sin(t) u(t)$$

2b

$$V_{out} = V_s(t) \left(\frac{j\omega L}{R + j\omega L} \right)$$

$$\frac{V_{out}}{V_s(t)} = \hat{H}(\omega) = \frac{j\omega L}{R + j\omega L}, \quad |\hat{H}(\omega)| = \frac{\omega L}{R^2 + (\omega L)^2}$$

$$\angle = \tan^{-1}\left(\frac{\omega L}{R}\right)$$

$$\angle_{\text{numerator}} = \tan^{-1}\left(\frac{\omega L}{R}\right) = \tan^{-1}(\omega) = \frac{\pi}{2}$$

$$\rightarrow \sin(\pi/2) = 1, \cos(\pi/2) = 0$$

$$\rightarrow \tan(\pi/2) = (1/0) = \infty$$

$$\angle_{\text{denominator}} = \tan^{-1}\left(\frac{\omega L}{R}\right)$$

$$\text{Phase} = \phi(\omega) = \angle_{\text{numerator}} - \angle_{\text{denominator}}$$

$$\phi(\omega) = \frac{\pi}{2} - \tan^{-1}\left(\frac{\omega L}{R}\right)$$

$$\omega_0 (\text{Fundamental Frequency}) = \frac{2\pi}{T_0}$$

$$T_0 (\text{Period}) = \text{End of Cycle} - \text{Beginning of Cycle} = \frac{1}{2} - \left(-\frac{1}{2}\right) = \frac{1}{1} = T_0$$

$$\omega_0 = \frac{2\pi}{T_0}$$

Sure. The key idea is that for a **linear system**, if the input is a sum of sinusoids, the output is the **same sum** but each term gets scaled by $|H(\omega)|$ and shifted by $\phi(\omega)$.

$$x(t) = \sum_{n=1, n=\text{odd}}^{\infty} \frac{8A}{n^2 \pi^2} \sin\left(\frac{n\pi}{2}\right) \sin\left(\frac{2n\pi t}{T_0}\right) \rightarrow \sin\left(\frac{2\pi}{T_0} t\right) = \sin(\omega_0 t)$$

$$\rightarrow y(t) = \sum_{n=1, n=\text{odd}}^{\infty} \frac{8A}{n^2 \pi^2} \times |H(\omega)| \sin\left(\frac{n\pi}{2}\right) \sin\left(\frac{2n\pi t}{T_0} - \phi(\omega)\right)$$

$$= \sum_{n=1, n=\text{odd}}^{\infty} \frac{8A}{n^2 \pi^2} \times \frac{\omega_0 n L}{R^2 + \omega_0^2 n^2 L^2} \sin\left(\frac{n\pi}{2}\right) \sin\left(\frac{2n\pi t}{T_0} + \frac{\pi}{2} - \tan^{-1}\left(\frac{\omega_0 n L}{R}\right)\right)$$

Poles, Stability & LTI Systems 2.a) 1

The One Rule
All poles of $H(s)$ must have strictly negative real parts to be stable.

Finding Poles
Set each denominator factor to zero and solve for s .

Factor	Roots
$s+3$	$s=-3$
s^2-4	$s=\pm 2$
s^2+4	$s=\pm j2$
$(s+1)^2+1$	$s=-1\pm j$
s^3+5s+4	$s=-1, -4$

The j Rule 2.a)

$\pm j1 \Rightarrow 1$ by definition. Whenever you take the square root of a negative number you get j , meaning the root is purely imaginary $\rightarrow$ lands on the imaginary axis $\rightarrow$ unstable.

BIBO Stability 3

Bounded Input Bounded Output — a property of the system only:

- Only check $H(s)$ (the system transfer function) for stability
- Ignore $X(s)$ (the input) and $Y(s)$ (the output)
- $Y(s) = H(s) \cdot X(s)$, but the $1/s^2$ from a ramp input is irrelevant to whether the system itself is stable

Example $h(t) = e^{-t} \cos(t) u(t)$

$H(s) = (s+1) / ((s+1)^2 + 1) \rightarrow$ poles at $-1 \pm j \rightarrow$ real part = $-1 \rightarrow$ BIBO stable

The e^{-t} decay in $h(t)$ is the time-domain giveaway that poles are in the left half plane

Quick Checks

- Linear factor $(s+a) \rightarrow$ one real pole at $s=-a$. Stable if $a > 0$
- Quadratic with no middle term $\rightarrow$ check sign $s^2 + 4$ forces imaginary, $s^2 - 4$ gives real roots
- Double pole at origin $(1/s^2) \rightarrow s=0$ twice $\rightarrow$ unstable, output grows as a ramp
- e^{-at} in $h(t) \rightarrow$ poles have negative real part $\rightarrow$ stable signal

The Repeated Root Rule 2.a) 5

If denominator has s^k , need every power from 1 up to k :

$$\frac{A}{s} + \frac{B}{s^2} + \dots + \frac{N}{s^k}$$

Common Mistake

Splitting the numerator is not partial fractions — the denominator stays the same and nothing simplifies. Always split by denominator factors.

Transform Pairs to Know

s-domain	Time-domain
A/s	$Au(t)$
B/s^2	$Btu(t)$
$b/((s+a)^2+b^2)$	$e^{-at} \sin(bt) u(t)$
$(s+a)/((s+a)^2+b^2)$	$e^{-at} \cos(bt) u(t)$

Why Cos Can Become Sine

The cos pair needs $(s+a)$ in the numerator. After partial fractions, if the numerator resolves to just a constant, it matches the sine pair instead. The change happens purely from algebra — no integration involved.

BIBO Stability Check

Only check $H(s)$ for stability — never $X(s)$ or $Y(s)$.

- e^{-at} in $h(t) \rightarrow$ poles have negative real part $\rightarrow$ stable
- Poles of $H(s)$ at $-1 \pm j \rightarrow$ real part = $-1 \rightarrow$ BIBO stable

The Three Outcomes 2.a) 2

Pole Location	Example	Verdict
Negative real part	$s = -2$ or $-1 \pm j$	Stable
Positive real part	$s = +2$	Unstable
Imaginary axis	$s = \pm j2$ or $s = 0$	Unstable

The $(s+a)^2 + b^2$ Rule

$(s+a)^2 + b^2 \rightarrow$ poles at $s = -a \pm jb$

- a controls the horizontal shift (real part)
- b controls the imaginary part
- $s^2 + 4$ is just $(s+0)^2 + 4 \rightarrow$ poles at $\pm j2 \rightarrow$ no shift $\rightarrow$ unstable

Factor	Poles	Stable?
$(s+1)^2 + 1$	$-1 \pm j$	✓
$(s+3)^2 + 4$	$-3 \pm j2$	✓
$(s-2)^2 + 4$	$+2 \pm j2$	✗
$s^2 + 4$	$\pm j2$	✗

Partial Fractions & Laplace Response — Study Notes

The Pipeline 2.a) 4

To find the output $y(t)$ given $h(t)$ and $x(t)$:

- Take Laplace transform of $h(t) \rightarrow H(s)$
- Take Laplace transform of $x(t) \rightarrow X(s)$
- Multiply: $Y(s) = H(s) \cdot X(s)$
- Partial fraction decompose $Y(s)$
- Inverse Laplace each term $\rightarrow y(t)$

No convolution needed — multiplication in s -domain replaces it.

Setting Up Partial Fractions

Each denominator factor gets its own term. The numerator degree must be one less than the denominator factor's degree (degree = highest power of s).

Denominator Factor	Term	Why
$(s+a)$	$A/(s+a)$	degree 1 $\rightarrow$ constant numerator
s^k repeated root	$A_k/s^k + B_k/s^{k-1} + \dots$	need one term per power
$(s+a)^2 + b^2$	$(Cs+D)/((s+a)^2+b^2)$	degree 2 $\rightarrow$ degree 1 numerator

3. (10 points) Laplace Transform & Its Inverse

a. (5 points) Derive the Laplace transform of $x(t) = A e^{-t} \sin(\omega_0 t + \theta) u(t)$.

What are the initial and final values of this signal?

$$\sin(\omega_0 t + \theta) = \sin(\omega_0 t) \cos \theta + \cos(\omega_0 t) \sin \theta$$

$$\mathcal{L}[\sin \omega_0 t u(t)] = \frac{\omega_0}{s^2 + \omega_0^2}$$

$$\mathcal{L}[\cos \omega_0 t u(t)] = \frac{s}{s^2 + \omega_0^2}$$

Therefore,

$$\sin(\omega_0 t + \theta) u(t) \xrightarrow{\mathcal{L}} \frac{\omega_0}{s^2 + \omega_0^2} \cos \theta + \frac{s}{s^2 + \omega_0^2} \sin \theta$$

$$A \sin(\omega_0 t + \theta) u(t) \xrightarrow{\mathcal{L}} \frac{A \omega_0}{s^2 + \omega_0^2} \cos \theta + \frac{A s}{s^2 + \omega_0^2} \sin \theta \quad \text{--- (A)}$$

Using the Frequency Shifting property of Laplace transforms, replace s by $(s+1)$

$$A e^{-t} \sin(\omega_0 t + \theta) u(t) \xrightarrow{\mathcal{L}} A \frac{\omega_0}{(s+1)^2 + \omega_0^2} \cos \theta + A \frac{(s+1)}{(s+1)^2 + \omega_0^2} \sin \theta \quad \text{--- (B)}$$

$\hookrightarrow e^{-at} x(t) \leftrightarrow X(s+a)$; put in place of s all s 's
 $e^{-t} = e^{-1t} \rightarrow (s+1)$

Initial value is obtained by substituting $t = 0+$, $x(0+) = A \sin(\theta)$ --- (C)

Final value is obtained as $t \rightarrow \infty$. Due to the exponential decay term, $x(\infty) = 0$. --- (D)

1 point for trying. 1 point for A-D

b. (5 points) Determine the inverse Laplace Transform of $H(s) = \frac{s^3 + 7s^2 + 4}{s^2 + 6s + 10}$.

1 point for trying. 1 point for A-D

Order of Nr > Order of Dr. Performing long division,

$$H(s) = \frac{s^3 + 7s^2 + 4}{s^2 + 6s + 10} = (s+1) - \frac{(16s+6)}{s^2 + 6s + 10} \quad \text{--- (A)}$$

$$= (s+1) - \frac{(16s+6)}{(s+3)^2 + 1} \quad \text{--- (B)}$$

$$= (s+1) - \frac{(16s+48-48+6)}{(s+3)^2 + 1}$$

$$= (s+1) - \frac{(16(s+3)-42)}{(s+3)^2 + 1}$$

$$= (s+1) - 16 \frac{(s+3)}{(s+3)^2 + 1} + 42 \frac{1}{(s+3)^2 + 1}$$

The inverse Laplace transform is

$$\frac{d}{dt} \delta(t) + \delta(t) - 16 e^{-3t} \cos(t) + 42 e^{-3t} \sin(t)$$

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3.a

$A e^{-t} \sin(\omega_0 t + \theta) u(t)$
not directly in laplace table, rewrite it to get into familiar form

Sum Identity $\rightarrow \sin(a+b) = \sin(a)\cos(b) - \cos(a)\sin(b)$

$$\sin(\omega_0 t + \theta) \rightarrow \sin(\omega_0 t) \cos(\theta) - \cos(\omega_0 t) \sin(\theta)$$

$$(\sin(\omega_0 t) \cos(\theta) - \cos(\omega_0 t) \sin(\theta)) u(t)$$

$$\rightarrow \sin(\omega_0 t) u(t) \cos(\theta) - \cos(\omega_0 t) u(t) \sin(\theta)$$

$$\rightarrow \mathcal{L} \left[\frac{\omega_0 \cos(\theta)}{s^2 + \omega_0^2} - \frac{s \sin(\theta)}{s^2 + \omega_0^2} \right]$$

$$\rightarrow \mathcal{L} [e^{-t}] = X(s+a) = (s+1) \rightarrow \text{replace where } s \text{ is}$$

$$A \left(\frac{\omega_0 \cos(\theta)}{(s+1)^2 + \omega_0^2} - \frac{s \sin(\theta)}{(s+1)^2 + \omega_0^2} \right)$$

$$A \frac{\omega_0 \cos(\theta)}{(s+1)^2 + \omega_0^2} - A \frac{s \sin(\theta)}{(s+1)^2 + \omega_0^2}$$

3.b

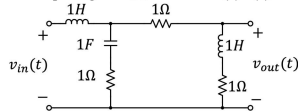
$$H(s) = \frac{s^3 + 7s^2 + 4}{s^2 + 6s + 10} \rightarrow \text{Numerator order} > \text{denominator order} \rightarrow \text{long division}$$

$$\frac{s^3 + 7s^2 + 4}{s^2 + 6s + 10} = s + 1 - \frac{16s + 6}{s^2 + 6s + 10}$$

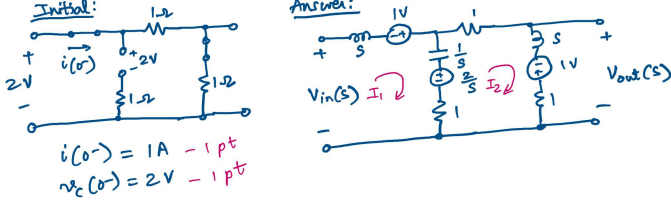
$1) \frac{s^3}{s^2} = s$
 $-s(s^2 + 6s + 10) = -s^3 - 6s^2 - 10s$
 $+8s^2 - 10s + 4$
 $2) \frac{8s^2}{s^2} = 8$
 $-8(s^2 + 6s + 10) = -8s^2 - 48s - 80$
 $+38s - 84$
 $3) \frac{38s}{s^2} = 38/s$
 $-38(s^2 + 6s + 10) = -38s^2 - 228s - 380$
 $+266s - 380$
 $4) \frac{266s}{s^2} = 266/s$
 $-266(s^2 + 6s + 10) = -266s^2 - 1596s - 2660$
 $+1330s - 2660$
 $5) \frac{1330s}{s^2} = 1330/s$
 $-1330(s^2 + 6s + 10) = -1330s^2 - 7980s - 13300$
 $+6650s - 13300$
 $6) \frac{6650s}{s^2} = 6650/s$
 $-6650(s^2 + 6s + 10) = -6650s^2 - 39900s - 66500$
 $+33250s - 66500$
 $7) \frac{33250s}{s^2} = 33250/s$
 $-33250(s^2 + 6s + 10) = -33250s^2 - 199500s - 332500$
 $+166250s - 332500$
 $8) \frac{166250s}{s^2} = 166250/s$
 $-166250(s^2 + 6s + 10) = -166250s^2 - 997500s - 1662500$
 $+831250s - 1662500$
 $9) \frac{831250s}{s^2} = 831250/s$
 $-831250(s^2 + 6s + 10) = -831250s^2 - 4987500s - 8312500$
 $+4156250s - 8312500$
 $10) \frac{4156250s}{s^2} = 4156250/s$
 $-4156250(s^2 + 6s + 10) = -4156250s^2 - 24937500s - 41562500$
 $+20781250s - 41562500$
 $11) \frac{20781250s}{s^2} = 20781250/s$
 $-20781250(s^2 + 6s + 10) = -20781250s^2 - 124687500s - 207812500$
 $+103906250s - 207812500$
 $12) \frac{103906250s}{s^2} = 103906250/s$
 $-103906250(s^2 + 6s + 10) = -103906250s^2 - 623437500s - 1039062500$
 $+519531250s - 1039062500$
 $13) \frac{519531250s}{s^2} = 519531250/s$
 $-519531250(s^2 + 6s + 10) = -519531250s^2 - 3117187500s - 5195312500$
 $+2597656250s - 5195312500$
 $14) \frac{2597656250s}{s^2} = 2597656250/s$
 $-2597656250(s^2 + 6s + 10) = -2597656250s^2 - 15585937500s - 25976562500$
 $+12988281250s - 25976562500$
 $15) \frac{12988281250s}{s^2} = 12988281250/s$
 $-12988281250(s^2 + 6s + 10) = -12988281250s^2 - 77929687500s - 129882812500$
 $+64941406250s - 129882812500$
 $16) \frac{64941406250s}{s^2} = 64941406250/s$
 $-64941406250(s^2 + 6s + 10) = -64941406250s^2 - 389648437500s - 649414062500$
 $+324707031250s - 649414062500$
 $17) \frac{324707031250s}{s^2} = 324707031250/s$
 $-324707031250(s^2 + 6s + 10) = -324707031250s^2 - 1948242187500s - 3247070312500$
 $+1623535156250s - 3247070312500$
 $18) \frac{1623535156250s}{s^2} = 1623535156250/s$
 $-1623535156250(s^2 + 6s + 10) = -1623535156250s^2 - 9741210937500s - 16235351562500$
 $+8117675781250s - 16235351562500$
 $19) \frac{8117675781250s}{s^2} = 8117675781250/s$
 $-8117675781250(s^2 + 6s + 10) = -8117675781250s^2 - 48706054687500s - 81176757812500$
 $+4054039187500s - 81176757812500$
 $20) \frac{4054039187500s}{s^2} = 4054039187500/s$
 $-4054039187500(s^2 + 6s + 10) = -4054039187500s^2 - 24324235125000s - 40540391875000$
 $+20250195937500s - 40540391875000$
 $21) \frac{20250195937500s}{s^2} = 20250195937500/s$
 $-20250195937500(s^2 + 6s + 10) = -20250195937500s^2 - 121501175625000s - 202501959375000$
 $+101250979687500s - 202501959375000$
 $22) \frac{101250979687500s}{s^2} = 101250979687500/s$
 $-101250979687500(s^2 + 6s + 10) = -101250979687500s^2 - 607505858125000s - 1012509796875000$
 $+506254879687500s - 1012509796875000$
 $23) \frac{506254879687500s}{s^2} = 506254879687500/s$
 $-506254879687500(s^2 + 6s + 10) = -506254879687500s^2 - 3037529278125000s - 5062548796875000$
 $+252502456250000s - 5062548796875000$
 $24) \frac{252502456250000s}{s^2} = 252502456250000/s$
 $-252502456250000(s^2 + 6s + 10) = -252502456250000s^2 - 1515014737500000s - 2525024562500000$
 $+1262512281250000s - 2525024562500000$
 $25) \frac{1262512281250000s}{s^2} = 1262512281250000/s$
 $-1262512281250000(s^2 + 6s + 10) = -1262512281250000s^2 - 7575073687500000s - 12625122812500000$
 $+6300061406250000s - 12625122812500000$
 $26) \frac{6300061406250000s}{s^2} = 6300061406250000/s$
 $-6300061406250000(s^2 + 6s + 10) = -6300061406250000s^2 - 37800368437500000s - 63000614062500000$
 $+31500307031250000s - 63000614062500000$
 $27) \frac{31500307031250000s}{s^2} = 31500307031250000/s$
 $-31500307031250000(s^2 + 6s + 10) = -31500307031250000s^2 - 189001842187500000s - 315003070312500000$
 $+157501535156250000s - 315003070312500000$
 $28) \frac{157501535156250000s}{s^2} = 157501535156250000/s$
 $-157501535156250000(s^2 + 6s + 10) = -157501535156250000s^2 - 945009210937500000s - 1575015351562500000$
 $+787507675781250000s - 1575015351562500000$
 $29) \frac{787507675781250000s}{s^2} = 787507675781250000/s$
 $-787507675781250000(s^2 + 6s + 10) = -787507675781250000s^2 - 4725046054687500000s - 7875076757812500000$
 $+3937538429687500000s - 7875076757812500000$
 $30) \frac{3937538429687500000s}{s^2} = 3937538429687500000/s$
 $-3937538429687500000(s^2 + 6s + 10) = -3937538429687500000s^2 - 23625230578125000000s - 39375384296875000000$
 $+19687692468750000000s - 39375384296875000000$
 $31) \frac{19687692468750000000s}{s^2} = 19687692468750000000/s$
 $-19687692468750000000(s^2 + 6s + 10) = -19687692468750000000s^2 - 118126154812500000000s - 196876924687500000000$
 $+162353969687500000000s - 196876924687500000000$
 $32) \frac{162353969687500000000s}{s^2} = 162353969687500000000/s$
 $-162353969687500000000(s^2 + 6s + 10) = -162353969687500000000s^2 - 974123818125000000000s - 1623539696875000000000$
 $+8117695351562500000000s - 1623539696875000000000$
 $33) \frac{8117695351562500000000s}{s^2} = 8117695351562500000000/s$
 $-8117695351562500000000(s^2 + 6s + 10) = -8117695351562500000000s^2 - 48706172109375000000000s - 81176953515625000000000$
 $+40540927968750000000000s - 81176953515625000000000$
 $34) \frac{40540927968750000000000s}{s^2} = 40540927968750000000000/s$
 $-40540927968750000000000(s^2 + 6s + 10) = -40540927968750000000000s^2 - 243245567812500000000000s - 405409279687500000000000$
 $+202504639687500000000000s - 405409279687500000000000$
 $35) \frac{202504639687500000000000s}{s^2} = 202504639687500000000000/s$
 $-202504639687500000000000(s^2 + 6s + 10) = -202504639687500000000000s^2 - 1215027837500000000000000s - 2025046396875000000000000$
 $+1662564250000000000000000s - 2025046396875000000000000$
 $36) \frac{1662564250000000000000000s}{s^2} = 1662564250000000000000000/s$
 $-1662564250000000000000000(s^2 + 6s + 10) = -1662564250000000000000000s^2 - 9975385500000000000000000s - 16625642500000000000000000$
 $+83125687500000000000000000s - 16625642500000000000000000$
 $37) \frac{83125687500000000000000000s}{s^2} = 83125687500000000000000000/s$
 $-83125687500000000000000000(s^2 + 6s + 10) = -83125687500000000000000000s^2 - 498754125000000000000000000s - 831256875000000000000000000$
 $+4156285000000000000000000000s - 831256875000000000000000000$
 $38) \frac{4156285000000000000000000000s}{s^2} = 4156285000000000000000000000/s$
 $-4156285000000000000000000000(s^2 + 6s + 10) = -4156285000000000000000000000s^2 - 24937710000000000000000000000s - 41562850000000000000000000000$
 $+207813250000000000000000000000s - 415628500000$

4. (20 points) Applications of the Laplace Transform

Consider the following circuit and the input signal $x(t) = 2 + \sin(t)u(t)$ V.



- a. (2 points) Sketch the Laplace-domain version of the given circuit with the initial conditions indicated (use the back sheet for your work as needed and write the final answer below).



- b. (3 points) Write the Laplace transform of the input for $t \geq 0$ (use the back sheet for your work and write the final answer below).

$$V_{in}(s) = \frac{1}{s^2+1}$$

- 1 pt for trying
 - 3 pts correct answer

$\mathcal{L}\{2 + \sin(t)u(t)\}$
 constant $\sim$ goes away

- c. (5 points) Calculate the Laplace-domain voltage $V_{out}(s)$ for $t \geq 0$. (use the back sheet for your work and write the final answer below). Your answer should be in the form of a ratio of two polynomials.

KVL:

$$\left\{ \begin{aligned} V_{in}(s) + 1 - \frac{2}{s} &= I_1(s + \frac{1}{s}) - I_2(1 + \frac{1}{s}) \\ \frac{2}{s} + 1 &= -I_1(1 + \frac{1}{s}) + I_2(1 + \frac{1}{s} + 1 + s + 1) \end{aligned} \right\}$$

$V_{out} = I_2(s+1) - 1$

$$\Delta = \frac{s^3 + 4s^2 + 4s + 2}{s} \Rightarrow I_2 = \frac{s^2 + 4s + 2 + (s+1)V_{in}}{s^3 + 4s^2 + 4s + 2}$$

$$\Delta_2 = \frac{s^2 + 4s + 2 + (s+1)V_{in}}{s} \Rightarrow V_{out} = \frac{(s^2 + 4s + 2 + (s+1)V_{in})s + 1}{s^3 + 4s^2 + 4s + 2} - 1$$

- d. An LTI system has the following differential equation description, with initial conditions $x(0^-) = 1$, $y(0^-) = 1$, $x'(0^-) = 1$, $y'(0^-) = 1$, and input $x(t) = te^{-2t}u(t)$:

$$s^2 + 12s + 35 = s + 3$$

$$\frac{d^2y}{dt^2} + 12\frac{dy}{dt} + 35y = \frac{dx}{dt} + 3x$$

- (i) (5 points) Identify the poles, zeros, and modes of this system.

$s^2 + 12s + 35 = (s+5)(s+7)$ characteristic polynomial

$$H(s) = \frac{(s+3)}{(s+5)(s+7)}$$

poles: $\{-5, -7\}$ (A)
 zero: $\{-3\}$ (B)
 modes: $\{-5, -7\}$ (C)

grading: 1 pt trying, 2 pts (A, B, C), 1 pt each.

- (ii) (5 points) Calculate the total response of this system in the Laplace domain.

+ 2 trying

$$[s^2 Y(s) - s y(0^-) - y'(0^-)] + [12(s Y(s) - y(0^-))] + 35 Y(s) = [s X(s) - x(0^-)] + 3 X(s)$$

$$\Rightarrow [s^2 Y(s) - s(1) - (1)] + [12(s Y(s) - 1)] + 35 Y(s) = s X(s) - 1 + 3 X(s)$$

$$\Rightarrow s^2 Y(s) - s - 1 + 12s Y(s) - 12 + 35 Y(s) = s X(s) - 1 + 3 X(s)$$

$$\Rightarrow s^2 Y(s) + 12s Y(s) + 35 Y(s) = X(s)(s+3) + 12 + s$$

$$\Rightarrow Y(s)[s^2 + 12s + 35] = X(s)(s+3) + 12 + s$$

$$\Rightarrow X(s) = \mathcal{L}\{x(t)\} = \mathcal{L}\{te^{-2t}u(t)\} = \frac{1}{(s+2)^2}$$

$$\Rightarrow Y(s)[s^2 + 12s + 35] = \frac{1}{(s+2)^2} \cdot (s+3) + 12 + s$$

$$Y(s) = \frac{(s+3)}{(s+2)^2} + 12 + s$$

+ 1

5. (25 points) Fourier Series and Transform

- a. (7 points). Find the Fourier Series of the signal $x(t) = |\sin(t)|$ in complex exponential representation where $||$ indicates the absolute value.

+ 1

$$T_0(\sin(t)) = 2\pi$$

$$T_0(|\sin(t)|) = \pi$$

$$\omega_0 = \frac{2\pi}{T_0} = \frac{2\pi}{\pi} = 2$$

$$C_n = \frac{1}{T_0} \int_0^{T_0} x(t) e^{-jn\omega_0 t} dt$$

$$= \frac{1}{\pi} \int_0^{\pi} (\sin(t)) e^{-j2nt} dt$$

Euler's Formula $\rightarrow \sin(t) = \frac{e^{jt} - e^{-jt}}{2j}$

$$\rightarrow \frac{1}{\pi} \int_0^{\pi} \frac{e^{jt} - e^{-jt}}{2j} e^{-j2nt} dt \rightarrow \frac{1}{2j\pi} \int_0^{\pi} (e^{jt} - e^{-jt}) e^{-j2nt} dt$$

$$\rightarrow \frac{1}{2j\pi} \int_0^{\pi} e^{jt} e^{-j2nt} dt - \frac{1}{2j\pi} \int_0^{\pi} e^{-jt} e^{-j2nt} dt$$

$$\rightarrow \frac{1}{2j\pi} \int_0^{\pi} e^{j(1-2n)t} dt - \frac{1}{2j\pi} \int_0^{\pi} e^{-j(1+2n)t} dt$$

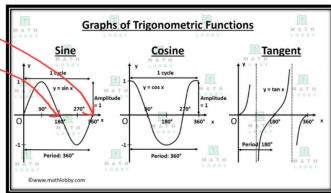
$$\rightarrow \frac{1}{2j\pi} \left(\frac{e^{j\pi(1-2n)}}{j(1-2n)} - \frac{e^{j\pi(1+2n)}}{j(1+2n)} \right) \Big|_0^{\pi}$$

$$\rightarrow \frac{1}{2j\pi} \left[\left(\frac{e^{j\pi(1-2n)}}{j(1-2n)} - \frac{1}{j(1-2n)} \right) - \left(\frac{e^{j\pi(1+2n)}}{j(1+2n)} - \frac{1}{j(1+2n)} \right) \right]$$

$$\rightarrow \frac{1}{2j\pi} \left(\frac{e^{j\pi(1-2n)} - 1}{j(1-2n)} - \frac{e^{j\pi(1+2n)} - 1}{j(1+2n)} \right) \rightarrow \frac{1}{2j\pi} \left(\frac{e^{j\pi(1-2n)} - 1}{j(1-2n)} + \frac{e^{-j\pi(1+2n)} - 1}{j(1+2n)} \right)$$

$$\rightarrow e^{j\pi(1-2n)} \& e^{-j\pi(1+2n)} = -1$$

+ 1



Recognizing Exponential Patterns — Study Notes

The Core Rule

For $e^{jk(n)}$ where k is an integer:

- k odd $\rightarrow e^{jk(n)} = -1$
- k even $\rightarrow e^{jk(n)} = +1$

Forms to Recognize

Always odd $\rightarrow$ always -1 :

- odd $\pm 2n$ (e.g. $1 \pm 2n, 3 \pm 2n$)
- Rule: odd $\pm$ even = always odd

Always even $\rightarrow$ always $+1$:

- even $\pm 2n$ (e.g. $2 \pm 2n, 4 \pm 2n$)
- Rule: even $\pm$ even = always even

Could be either $\rightarrow$ test values:

- Any time n is NOT multiplied by an even number
- e.g. just $n, n+3, 3n$
- These flip between odd and even depending on what n is

The Quick Check

Just look at the constant term (the part without n) and whether n is multiplied by an even number:

- Constant is odd $\pm n$ multiplied by even $\rightarrow$ always -1
- Constant is even $\pm n$ multiplied by even $\rightarrow$ always $+1$
- Anything else $\rightarrow$ test $n = 1, 2, 3$ and see

Yes — it all traces back to Euler's formula:

$$e^{jn\pi} = \cos(n\pi) + j\sin(n\pi)$$

- $\sin(n\pi) = 0$ always for any integer n — so the imaginary part always vanishes

- $\cos(n\pi)$ is what determines the result — it alternates between $+1$ and -1 depending on whether n is even or odd

So the odd/even rules are really just saying what $\cos(n\pi)$ evaluates to, which comes directly from Euler's formula. Without Euler's formula you'd have no way to simplify $e^{j(n\pi)}$ into just $+1$ or -1 .

$$\rightarrow \frac{1}{2j^2\pi} \left[\frac{-1-1}{(1-2n)} + \frac{-1-1}{(1+2n)} \right] \rightarrow \frac{1}{2j^2\pi} \left[\frac{-2}{(1-2n)} + \frac{-2}{(1+2n)} \right]$$

$$\rightarrow j^2 = -1$$

$$\rightarrow \frac{1}{2\pi} \left[\frac{2}{(1-2n)} + \frac{2}{(1+2n)} \right] \rightarrow \frac{1}{2\pi} \left[\frac{1}{(1-2n)} + \frac{1}{(1+2n)} \right]$$

$$\rightarrow (a+b)(a-b) = (a^2 - b^2)$$

$$\rightarrow (1-2n)(1+2n) = (1^2 - 4n^2)$$

$$\rightarrow \frac{1}{2\pi} \left[\frac{2}{(1^2 - 4n^2)} \right] \rightarrow \frac{2}{\pi} \cdot \frac{1}{(1^2 - 4n^2)} = X_n$$

+ 1

- b. (6 points) The signal $x_2(t)$ is defined as follows

$$x_2(t) = \sum_{n=0}^{\infty} e^{-jn(3\omega_0+2)} \sin(2n\pi\omega_0) e^{jn\omega_0 t}$$

Find the Fourier Series of $x_2(t)$ in cosine/sine form. Please write your final answer below:

$$a_0 = \frac{0}{1}$$

$$a_n = \frac{e^{-jn(3\omega_0+2)} \sin(2n\pi\omega_0)}{1}$$

$$b_n = \frac{e^{-jn(3\omega_0+2)} \sin(2n\pi\omega_0)}{1}$$

$$x_2(t) = \sum_{n=0}^{\infty} e^{-jn(3\omega_0+2)} \sin(2n\pi\omega_0) e^{jn\omega_0 t}$$

Substituting $n = 0$ gives the DC term.

$$a_0 = 0 \quad \text{(A)}$$

Using Euler's identifies, $e^{jn\omega_0 t} = \cos(n\omega_0 t) + j \sin(n\omega_0 t)$. Therefore,

$$x_2(t) = a_0 + \sum_{n=1}^{\infty} e^{-jn(3\omega_0+2)} \sin(2n\pi\omega_0) e^{jn\omega_0 t}$$

$$= a_0 + \sum_{n=1}^{\infty} e^{-jn(3\omega_0+2)} \sin(2n\pi\omega_0) [\cos(n\omega_0 t) + j \sin(n\omega_0 t)]$$

$$= a_0 + \sum_{n=1}^{\infty} e^{-jn(3\omega_0+2)} \sin(2n\pi\omega_0) \cos(n\omega_0 t) + e^{-jn(3\omega_0+2)} \sin(2n\pi\omega_0) j \sin(n\omega_0 t)$$

We can identify a_n and b_n from the above expression.

$$a_n = e^{-jn(3\omega_0+2)} \sin(2n\pi\omega_0) \quad \text{--- (B)}$$

$$b_n = j e^{-jn(3\omega_0+2)} \sin(2n\pi\omega_0) \quad \text{--- (C)}$$

1 point for trying. 2 points each for A-C

- c. (6 points) The Fourier transform of a signal $x(t)$ is $X(\omega)$. Find the Fourier transform of the signal $y(t) = x(2t) * x(-3t - 3)$, where $*$ represents convolution.

1 point for trying. 1 point for A-E

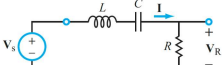
$$x(t) \xrightarrow{\mathcal{F}} X(\omega)$$

$$x(2t) \xrightarrow{\mathcal{F}} \frac{1}{2} X\left(\frac{\omega}{2}\right) \quad \text{--- (A)}$$

(Table 5-7, #3 Time scaling property).

6. (20 points) Applications of Fourier Methods

- a. (5 points) Write down the transfer function of the following circuit in the frequency domain.



$$H(\omega) = \frac{j\omega RC}{(1-\omega^2 LC) + j\omega RC}$$

• 1 pt trying
• 5 pts correct answer

- b. (5 points) Calculate the center frequency ω_0 , bandwidth B , and the frequency ω_{45} at which the phase shift of the filter in part (a) above is 45° , when $L = 2H$, $C = 1F$, and $R = 1\Omega$.

$$\text{Center frequency } \omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{2 \times 1}} = \frac{1}{\sqrt{2}} \text{ rad/sec} \quad \text{(A)}$$

$$B = \text{Bandwidth} = \frac{R}{L} = \frac{1}{2} \text{ rad/sec} \quad \text{(B)}$$

$$\phi(\omega) = \angle H(\omega) = \frac{\pi}{2} - \tan^{-1}\left(\frac{\omega RC}{1-\omega^2 LC}\right) = \frac{\pi}{4}$$

$$\Rightarrow \omega RC = 1 - \omega^2 LC \Rightarrow \omega^2 LC + \omega RC - 1 = 0$$

$$\Rightarrow 2\omega^2 + \omega - 1 = 0$$

$$\Rightarrow \text{Roots: } \frac{-1 \pm \sqrt{1^2 + 4 \cdot 2}}{4} \quad \text{Pick the positive root}$$

$$\omega_{45} = \frac{\sqrt{9} - 1}{4} = \frac{1}{2} \text{ rad/sec} \quad \text{(C)}$$

Grading:
• 1 pt trying
• +1 pt each
• (A) & (B)
• +2 pts (C)

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x(-3t) \xrightarrow{\mathcal{F}} \frac{1}{3} X\left(-\frac{\omega}{3}\right) \quad \text{--- (B)}$$

(Table 5-7, #3 Time scaling property).

$$x(-3(t+1)) \xrightarrow{\mathcal{F}} \frac{1}{3} X\left(-\frac{\omega}{3}\right) e^{j\omega} \quad \text{--- (C)}$$

(Table 5-7, #4 Time shifting property).

Convolution in the time domain results in multiplication in the Fourier domain (Table 5-7, #12)

$$Y(\omega) = \frac{1}{2} X\left(\frac{\omega}{2}\right) \cdot \frac{1}{3} X\left(-\frac{\omega}{3}\right) e^{j\omega} = \frac{1}{6} \left[X\left(\frac{\omega}{2}\right) X\left(-\frac{\omega}{3}\right) \right] e^{j\omega} \quad \text{--- (E)}$$

- d. (6 points) Find the Fourier transform of $x_4(t) = \text{sinc}(t - 10)e^{j100t}$. What kind of filter does $x_4(t)$ represent and why?

$$\text{Hint: } \text{sinc}(t) \xrightarrow{\mathcal{F}} \pi \text{rect}\left(\frac{\omega}{2}\right)$$

$$\text{sinc}(t) \xrightarrow{\mathcal{F}} \pi \text{rect}\left(\frac{\omega}{2}\right)$$

Using the time shifting property

$$\text{sinc}(t - 10) \xrightarrow{\mathcal{F}} \pi \text{rect}\left(\frac{\omega}{2}\right) e^{-j10\omega} \quad \text{--- (A)}$$

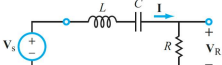
Using the frequency shifting property

$$\text{sinc}(t - 10) e^{j100t} \xrightarrow{\mathcal{F}} \pi \text{rect}\left(\frac{\omega - 100}{2}\right) e^{-j10(\omega - 100)}$$

This frequency spectrum is a rect function centered around $\omega = 100$ and width 2. Therefore, it represents a bandpass filter. (D)

1 point for trying. 1 point for A-E

- a. (5 points) Write down the transfer function of the following circuit in the frequency domain.



$$H(\omega) = \frac{j\omega RC}{(1-\omega^2 LC) + j\omega RC}$$

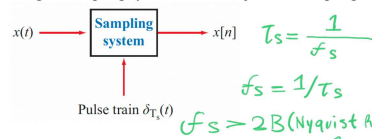
• 1 pt trying
• 5 pts correct answer

- c. (10 points) If the signal $x(t)$ is bandlimited to 10kHz, calculate the minimum sampling rate for the signal $y(t) = x^2(t)$ using the following available components that will ensure the sampled signal can be reconstructed perfectly.

- Low-pass filters with the following frequency response (B is adjustable):

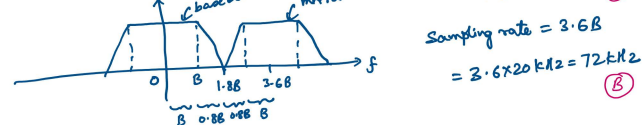


- A signal sampling system with an adjustable sampling rate $f_s = 1/T_s$.



Sketch a diagram showing how these components will be adjusted and interconnected.

$y(t)$ is bandlimited to 20kHz since multiplication in the time domain is equivalent to convolution in the frequency domain. $\Rightarrow B = 20\text{kHz}$ (A)



Diagram

